

Measures of Central Tendency

Introduction

Measures of central tendency are statistical values that describe the center or typical value of a dataset. The three main measures are the **mean**, **median**, and **mode**. Understanding when and how to use each measure is essential for accurately describing and analyzing data.

1. The Mean (Average)

Definition

The **mean** is the arithmetic average of all values in a dataset.

Formula

$$\begin{aligned}\text{Mean } (\bar{x}) &= (\text{Sum of all values}) / (\text{Number of values}) \\ &= (x_1 + x_2 + \dots + x_n) / n\end{aligned}$$

Example

Calculate the mean of: 4, 8, 6, 5, 3, 7

$$\begin{aligned}\text{Mean} &= (4 + 8 + 6 + 5 + 3 + 7) / 6 \\ &= 33 / 6 \\ &= 5.5\end{aligned}$$

###When to Use

-  When data is **normally distributed** (bell curve)
-  When you need to use **all data values**
-  For **interval or ratio data**

When NOT to Use

-  When data has **outliers** (extreme values)
-  When data is **skewed** (not symmetric)
-  With **ordinal or categorical data**

Key Properties

- Uses every data value in its calculation
 - Sensitive to extreme values (outliers)
 - Can be a value that doesn't exist in the dataset
 - Most commonly used measure
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2. The Median

Definition

The **median** is the middle value when data is arranged in order. Half the values are above it, half are below.

How to Calculate

For **ODD** number of values:

1. Arrange data in ascending order
2. Find the middle value

For **EVEN** number of values:

1. Arrange data in ascending order
2. Find the two middle values

3. Calculate their average

Example 1 (Odd): Find median of: 12, 15, 8, 19, 22, 10, 13

Step 1: Order the data

8, 10, 12, 13, 15, 19, 22

Step 2: Find middle value (position 4)

Median = 13

Example 2 (Even): Find median of: 2, 5, 7, 10

Step 1: Order the data

2, 5, 7, 10

Step 2: Find two middle values

Middle values: 5 and 7

Step 3: Calculate average

Median = $(5 + 7) / 2 = 6$

When to Use

- ☒ When data has **outliers**
- ☒ When data is **skewed**
- ☒ With **ordinal data** (ranked categories)
- ☒ When you need a **resistant measure**

Key Properties

- Resistant to outliers
- Only depends on position, not actual values
- Always a real data value (for odd n) or average of two values (for even n)
- Better represents "typical" value in skewed distributions

3. The Mode

Definition

The **mode** is the value that appears most frequently in a dataset.

Types of Distributions by Mode

- **Unimodal:** One mode
- **Bimodal:** Two modes
- **Multimodal:** More than two modes
- **No mode:** All values occur equally (or all unique)

Example 1 (Unimodal)

Dataset: 5, 8, 8, 12, 15, 8, 20

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Value 8 appears 3 times (most frequent)
Mode = 8
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Example 2 (Bimodal)

Dataset: 2, 3, 3, 4, 5, 5, 6

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Values 3 and 5 both appear twice
Modes = 3 and 5 (bimodal)
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Example 3 (No Mode)

Dataset: 1, 2, 3, 4, 5

All values appear once
No mode

When to Use

- ☒ With **categorical data** (colors, categories)
- ☒ To find the **most common value**
- ☒ When frequency matters more than magnitude
- ☒ With **any type of data**

Key Properties

- Only measure that works with categorical data
- Can have multiple modes or no mode
- Not affected by extreme values
- May not be near the center of the distribution

4. Comparing Mean, Median, and Mode

Relationship in Different Distributions

Symmetric Distribution (Normal)

Mean = Median = Mode

All three measures give the same value.

Right-Skewed Distribution

Mode < Median < Mean

The mean is pulled toward the high values (right tail).

Left-Skewed Distribution

Mean < Median < Mode

The mean is pulled toward the low values (left tail).

Quick Reference Table

Feature	Mean	Median	Mode
Formula	Sum ÷ Count	Middle value	Most frequent
Data Type	Numerical only	Ordinal/Numerical	Any type
Affected by outliers	Yes	No	No
Unique value	Always	Always	Not always
Uses all data	Yes	No	No

5. Real-World Applications

Scenario 1: Test Scores

Data: 85, 88, 90, 92, 95, 97, 98

- **Best measure:** Mean (data is normally distributed, no outliers)
- Mean = 92.1 ← Use this!
- Median = 92
- Mode = None

Scenario 2: House Prices

Data: \$200K, \$220K, \$230K, \$240K, \$2.5M

- **Best measure:** Median (extreme outlier: \$2.5M mansion)
- Mean = \$678K ← Misleading!
- Median = \$230K ← Better representation
- Mode = None

Scenario 3: Shoe Sizes Sold

Data: 7, 7, 8, 8, 8, 9, 9, 10

- **Best measure:** Mode (finding most common size)
- Mean = 8.25 (not a real shoe size)
- Median = 8
- Mode = 8 ← Most useful!

6. Common Mistakes to Avoid

✗ Mistake 1: Using mean with skewed data

 **Problem:** Income data with billionaires

- Mean income = \$500,000 (misleading!)
- Median income = \$50,000 (better)

✗ Mistake 2: Forgetting to order data for median

Wrong: Find median of 5, 2, 9, 3

- Middle of 5, 2, 9, 3 = average of 2 and 9 = 5.5 ✗

Correct: Order first!

- 2, 3, 5, 9 → Median = $(3 + 5) / 2 = 4$ ✓

✗ Mistake 3: Assuming mode is always near center

Mode can be anywhere in the distribution! It's about frequency, not position.

7. Practice Problems

Problem 1

Find mean, median, and mode of: 10, 15, 12, 10, 20, 18, 10

► [Click for solution](#)

Problem 2

Which measure is best for describing typical salary in a company with one CEO making \$5M and 100 employees making \$40K-\$60K?

► [Click for solution](#)

Key Takeaways

1. **Mean** = arithmetic average, uses all data, sensitive to outliers
 2. **Median** = middle value, resistant to outliers, best for skewed data
 3. **Mode** = most frequent value, works with any data type
 4. Choose the measure based on your data's **distribution** and **purpose**
 5. In symmetric distributions, all three measures are similar
 6. In skewed distributions, median is usually preferred
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Additional Resources

- Use the **Statistics Calculator** on our website to automatically calculate all three measures
 - Take the **Practice Quiz** to test your understanding
 - Download the **Statistical Tables Cheat Sheet** for quick reference
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Ready to test your knowledge? Take the interactive quiz to reinforce these concepts!

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